MANAS PANDEY

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PROFESSIONAL SUMMARY

Software Developer with 1+ years of experience building **Generative AI and LLM-based systems** in production – **LangChain/LangGraph** multi-agent workflows, **RAG** pipelines, Groq LLM integration, and prompt engineering – alongside scalable REST APIs using Python (**Django, Flask, FastAPI**).
Broader background spans full-stack development (**React.js**), computer-vision pipelines (**YOLOv8/OpenCV/OCR**), and strong database design (**MySQL, PostgreSQL, MongoDB, Redis**). Owns the full SDLC in Agile teams, with containerized (**Docker**) production deployments.
Broader engineering background spans computer-vision pipelines (**YOLOv8/OpenCV**), authentication services, and full-stack platforms, with strong SQL/MySQL schema design and OOP/DSA fundamentals. Owns the full SDLC in Agile teams: clean architecture, containerized deployment (**Docker**), and production-hardened APIs. Actively expanding into structured prompt evaluation and optimization workflows.

TECHNICAL SKILLS

- **Languages:** Python, SQL, JavaScript
- **AI & ML:** LangChain, LangGraph, Multi-Agent Systems, RAG Pipelines, Groq LLMs, TensorFlow, PyTorch, Computer Vision (YOLOv8, OpenCV, OCR), ETL Pipelines
- **Frameworks & Backend:** Django, Flask, FastAPI, REST API Design & Development, Microservice Architecture, Celery
- **Database:** MySQL, PostgreSQL, MongoDB, Redis
- **Dev Tools & Cloud:** Docker, Kubernetes, OpenShift, Git/GitHub, CI/CD, Jenkins, SonarQube, Checkmarx, AWS, GCP, Nginx, Linux (Ubuntu), Postman
- **Frontend:** React.js, Redux
- **Coursework:** Object-Oriented Programming, Data Structures & Algorithms, Software Design Principles, SDLC

WORK EXPERIENCE

Software Developer, iLogitek Business Solutions *May 2026 – October 2026*

- Designed and developed scalable backend services (FastAPI, Django, Flask) and REST APIs, applying OOP and software design principles across the full SDLC; built data-processing and automation modules integrated into full-stack product workflows serving live users.
- Managed relational and NoSQL databases (MySQL, PostgreSQL, MongoDB); implemented secure, production-hardened API services with structured error handling, unit testing, and Docker-based deployment.
- Collaborated with cross-functional teams in an Agile environment to translate business requirements into technical solutions; participated in client-facing presentations to communicate findings and technical decisions.

Software Developer, Vinayan India Consulting *May 2025 – Dec 2025*

- Built an edge-deployed vehicle e-challan system on **Intel NUC** hardware with a laser trigger sensor and camera, using **YOLOv8** and **OpenCV** for real-time vehicle detection and plate cropping, with **OCR** for plate extraction, reducing manual review effort by 40%; exposed captured violations via FastAPI REST endpoints backed by MongoDB for traffic authorities to review and process challans.
- Collaborated with cross-functional teams in an Agile environment, communicating technical decisions and progress to stakeholders; wrote unit tests and technical documentation to support maintainability.

Software Developer Intern, CRIS – Centre for Railway Information System *Jul 2024 – Sep 2024*

- Developed a freight demand forecasting model in Python; performed EDA, data cleaning, and feature engineering on railway freight datasets, then built FastAPI REST endpoints to expose outputs, collaborating with senior engineers on model integration and code review.

PROJECTS

1. Vehicle E-Challan System – Automated Number Plate Recognition (ANPR) *Python, FastAPI, MongoDB, OpenCV, YOLOv8, OCR, Ubuntu 22.04 LTS*

- Built an edge-deployed traffic violation detection system on **Intel NUC** hardware integrated with a laser trigger sensor (Dragon Eye) and camera, capturing vehicles in real time as they committed traffic violations.
- Used **YOLOv8** for vehicle detection and **OpenCV** for image preprocessing and plate cropping on captured frames, with **OCR** to extract license plate numbers from each violation.
- Exposed captured violation records via REST APIs (**FastAPI**) backed by **MongoDB**, enabling traffic authorities to manually review captured evidence and process challans.
- Deployed and ran the full pipeline on Ubuntu 22.04 LTS in a field edge-computing setup.

2. SimpleTech Training – Institute Management Platform *FastAPI, React/Vite, MySQL, Redis, Celery, Docker Compose, Nginx*

- Designed and developed a full-stack training institute management platform using FastAPI and React/Vite, covering student enrollment, batch scheduling, fee tracking, and course material delivery, following SDLC best practices end-to-end.
- Built and exposed REST APIs (FastAPI) for core platform operations, backed by a normalized MySQL relational schema for student, batch, and payment records, with input validation and structured error handling.
- Implemented asynchronous email workflows (forgot/reset password, notifications) using Celery and Redis to keep API response times fast under load.
- Containerized the full stack with Docker Compose and configured Nginx as a reverse proxy for production deployment; wrote unit tests to validate core enrollment and payment flows.

3. MindHeal – Mental Wellness Platform *React.js, FastAPI, MySQL, WebSocket, JWT, Google OAuth, Redis, AWS*

- Shipped a full-stack mental wellness platform using React.js and FastAPI with 3 distinct user roles (user, therapist, admin), supporting appointment booking, scheduling, and session management.
- Built WebSocket-based real-time chat scoped per appointment, and REST APIs backed by a MySQL relational schema for appointments, chat history, and payments.
- Implemented authentication with JWT and Google OAuth (including OTP verification), applying role-based access control across user, therapist, and admin flows.
- Deployed and production-hardened the platform on AWS/Railway with Redis for session and WebSocket management, resolving real-world issues around CORS, API routing, and persistent connections.

4. SecureAuthX – AI-Powered Authentication System *Python, Django/Flask, JWT, MySQL, REST API, ML*

- Built a reusable authentication system using Django/Flask with secure signup/login flows, password hashing, and JWT-based session handling, exposed via REST APIs for easy integration into other projects.
- Integrated an ML-based anomaly detection layer to flag suspicious login patterns (unusual location/device/time), triggering step-up verification for risk-flagged sessions.
- Implemented role-based access control (RBAC) to manage permissions across different user types, with token refresh and expiry handling for secure long-lived sessions.
- Designed the MySQL schema for users, roles, tokens, and login-risk logs, applying OOP and modular software design principles to keep the auth module framework-agnostic and reusable.

EDUCATION

Sunder Deep Engineering College (Affiliated to AKTU) *2021 – 2025*

Bachelor of Technology (CSE) *Ghaziabad, Uttar Pradesh*

HackerRank: 5-Star Python and SQL | **Training:** Ducat IT – Python, ML, Data Analysis

CERTIFICATIONS

- **Microsoft Certified: Azure AI Fundamentals (AI-901)** *Issued Jul 2026*
- TensorFlow, Deep Learning, NLP